# SHIVAM SINGH CHAUHAN

Generative AI Engineer | AIML Engineer

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## SUMMARY

B.Tech CSE graduate (2026) with a strong foundation in Python, Natural Language Processing (NLP), Transformers, and Generative AI. Built LLM-powered applications using LangChain, LangGraph, LlamaIndex, vector databases, and modern AI tools. Passionate about translating AI research into practical solutions and continuously exploring emerging AI technologies.

## TECHNICAL SKILLS

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|---|---|
| **Languages** | Python, SQL |
| **GenAI & LLM** | LangChain, LangGraph, AI Agents, Tool Calling, Structured Output (Pydantic), Prompt Engg. |
| **RAG** | Chunking Strategies, Hybrid Search (BM25 + Vector), Reranking, Query Rewriting, Metadata Filtering |
| **LLM Evaluation & Safety** | RAGAS, LLM-as-a-Judge, Guardrails AI, LangSmith (Tracing & Observability) |
| **NLP & ML** | Transformer Architecture, Tokenization, Embeddings, Hugging Face, NumPy, Pandas |
| **Vector DB & Databases** | ChromaDB, FAISS, Pinecone, SQLite, PostgreSQL, Redis |
| **Backend & Tools** | FastAPI, Streamlit, Docker, Git, GitHub, VS Code, Jupyter, Groq / OpenAI APIs |

## PROJECTS

### Self-Healing RAG 09/2026

*LangGraph, LangChain, ChromaDB, Hugging Face, Streamlit*

- Built a document Q&A system that **checks its own answers**: a critic verifies each answer against the retrieved text and, if it is unsupported, rewrites the query and retries in a **LangGraph** loop.
- Combined **ChromaDB** vector search with **BM25** keyword search and a **cross-encoder reranker**.
- Scored **96.6% answer accuracy** and **100% retrieval recall** on a hand-written 29-question test set.

### LLM Guardrails Gateway 07/2026

*FastAPI, LangGraph, Prompt-Guard-2, Redis, Pydantic*

- Built a **FastAPI** gateway that blocks **prompt injection, jailbreaks and PII**, and flags policy violations in outputs.
- Added Meta's **Prompt-Guard-2** to regex rules: attack recall **0.66** → **0.80** on 80 prompts, 3.3% false positives.
- Used **Redis** for per-key **rate limiting** and **caching guard decisions**, so repeated prompts skip the classifier.
- Runtime-editable YAML policies, audit logging and a **/metrics** endpoint for block rate and p50/p99 latency.

### Agentic RAG Chatbot 06/2026

*LangGraph, FAISS, Streamlit, PostgreSQL, LangSmith*

- Built a chatbot with **streaming replies** and **saved chat threads** using LangGraph's SQLite checkpointer.
- Turned it into a **tool-calling agent** that picks between web search, APIs and **uploaded PDFs** indexed in FAISS.
- Used **LangSmith** traces to debug the agent's tool choices and measured answer quality with **RAGAS**.

### Customer Feedback Classifier 12/2025

*LangChain (LCEL), LangGraph, Pydantic, FastAPI, SQLite, Streamlit*

- Classifies customer feedback into a typed **Pydantic** schema (sentiment, 8 categories, urgency, bounded confidence, summary, action items), served via FastAPI and a Streamlit dashboard.
- Built a **self-correcting LangGraph loop**: when the **PydanticOutputParser** rejects the LLM's output, the validation error is sent back so the model repairs its own JSON.
- Composed **LCEL** chains with rule-based few-shot prompts; a second chain turns stored results into a validated insights report (risks, wins, trends, actions).

## EDUCATION

**Greater Noida Institute of Technology (GNIOT)** 2022 – 2026
B.Tech, Computer Science & Engineering | **CGPA: 6.7/10** Greater Noida

**Saraswati Gyan Mandir Inter College** 2021 | 2019
Class XII: **88%** | Class X: **86.5%** Agra

## CERTIFICATIONS

**Advanced RAG** – CampusX 07 2026

**Data Science with Python** – E&ICT Academy, IIT Kanpur Jul – Aug 2025